

Comments: Domain Capabilities Profiles New York State Grid of the Future Touchpoint 3

May 26, 2026

Vehicle-Grid Integration Council (“VGIC”)¹ is a 501(c)(6) membership-based advocacy organization committed to advancing the role of vehicle-grid integration (“VGI”) through policy development, education, and research. VGIC supports the transition to decarbonized transportation and electric sectors by ensuring that the value from flexible electric vehicle (“EV”) charging and discharging is recognized, compensated, and aligned with a more reliable, affordable, and efficient electric grid.

1. Introduction

VGIC appreciates the opportunity to provide comments on the New York State Grid of the Future (“GOTF”) Domain Capability Profiles. VGIC commends the Department of Public Service (“DPS”), NYSERDA, and the broader GOTF team for undertaking an ambitious and thoughtful effort to develop a comprehensive framework to advance the evolution of New York’s electric power ecosystem.

The pre-read document for GOTF Touchpoint 3 (review of the Domain Capability Profiles) appropriately recognizes that enabling grid flexibility will be central to maintaining affordability and reliability while integrating increasing levels of electrification and renewable energy resources. Achieving this vision will require coordination across sectors and among electric sector stakeholders to a degree that has not yet been achieved in New York’s electric power ecosystem.

¹ VGIC member companies and supporters include American Honda Motor Co., Inc., Ford Motor Company, General Motors, Nissan Group of North America, Bidirectional Energy, BMW of North America LLC, bodeEV, ChargePoint, ChargeScape, Codibly Inc., Critical Loop, dcbe, Eaton Corporation, EnergyHub, EV.Energy, Fermata Energy, Gravity Inc., Mercedes-Benz Research & Development North America, Inc., Orange Charger, Inc., Renew Home, Rivian, Stellantis, SWITCH Energy Inc., Tesla, The Mobility House, Toyota Motor North America, Inc., UL Solutions, WeaveGrid, Clean Power Alliance, Connecticut Green Bank, Los Angeles Department of Water and Power, Marin Clean Energy, San Diego Community Power, and Sacramento Municipal Utility District. The views expressed in these Comments are those of VGIC, and do not necessarily reflect the views of all individual VGIC member companies or supporters. (<https://www.vgicouncil.org/>)

VGIC believes the linkage between the Grid Flexibility Study and the Domain Capability Profiles could be strengthened considerably. The pre-read document states that the Comprehensive Grid of the Future Plan builds directly upon the findings of the New York State Grid Flexibility Study and that the Study establishes “a common analytical framework to quantify how different types of flexible demand and supply resources could contribute to system needs over time.” The profiles do not yet fully incorporate the Grid Flexibility Study’s conclusion that EVs represent the single largest source of future grid flexibility potential in New York.

Transportation electrification represents both the largest source of future electric load growth and the largest opportunities to provide flexible demand and, with bidirectional charging systems, dispatchable storage. Electric vehicles are unique among distributed energy resources (“DER”) because they can simultaneously function as controllable load, dispatchable storage, backup power resources, and aggregated grid assets. As a result, EV flexibility should not be treated solely as a transportation electrification issue but rather reflected as a foundational planning and operational assumption throughout the domain capability profiles. This is particularly important given the emphasis on increasingly granular forecasting, flexibility-aware planning, DER orchestration, and co-optimized grid and DER investments across the domains.

2. Explicit Integration of Managed Charging and Bidirectional EVs as Dispatchable Grid Resources Across All Domains

VGIC supports the recognition within the domain capability profiles that the future electric system will require increasingly sophisticated approaches to forecasting, planning, interconnection, operations, and coordination across multiple domains. However, the capability profiles would benefit from more explicit treatment of electric vehicles and bidirectional charging systems as flexible grid resources capable of providing valuable grid services, rather than primarily characterizing transportation electrification as a source of future load growth. As EV adoption accelerates, managed charging and bidirectional charging technologies will increasingly play an important role in supporting grid flexibility, integrating renewable energy, reducing peak demand, and enhancing overall system reliability and resilience.

For example, within the Electricity Domain Capability Profile sub-capability Forecast Long-Term Distribution Load Growth (E1.2) appropriately references electrification forecasts, demand response participation estimates, time-based rates, and customer behavioral trends as key forecasting inputs. VGIC recommends that this section explicitly reference managed EV

charging behavior, bidirectional charging capability, and transportation electrification flexibility as distinct forecasting variables. The future operational characteristics of EV charging are fundamentally different from traditional static loads because charging behavior can increasingly respond to price signals, utility dispatch commands, operational constraints, renewable generation availability, and customer preferences.

Similarly, sub-capability Forecast Long-Term DER Generation Growth (E1.3) focuses primarily on traditional DER generation resources. VGIC recommends that this capability more explicitly recognize that bidirectional charging systems will increasingly function as dispatchable distributed storage resources capable of both importing and exporting energy under varying operational conditions. This distinction becomes increasingly important at higher maturity levels, where the framework begins discussing DER flexibility states, dispatch envelopes, curtailment potential, and flexibility-aware planning scenarios.

The Regulatory Domain Capability Profile similarly provides a strong foundation but could more clearly incorporate the role of EVs as flexible grid resources within future regulatory structures. VGIC supports the emphasis within the capability profile on integrating objectives across proceedings, coordinating regulatory actions, and developing adaptive frameworks for emerging technologies. Regulatory frameworks will increasingly need to recognize managed and bidirectional charging as important grid flexibility tools to meet the GOTF goals.

Sub-capability Develop Mechanisms to Regulate Emerging Technologies (R2.3) provides a relevant framework for bidirectional charging and VGI technologies. VGIC supports the recognition that regulatory frameworks for emerging technologies must “adapt quickly as new technologies evolve” and should “de-risk technologies” through coordinated efforts and stakeholder engagement. However, the capability profile could more explicitly reference VGI solutions as a priority emerging capability area requiring integrated regulatory treatment.

In addition to the above suggestions, VGIC encourages the GOTF drafting team to more explicitly incorporate VGI opportunities across the capability domain profiles. While the profiles appropriately emphasize flexibility, DER orchestration, distributed resource aggregation, and flexible load management, references to managed charging and bidirectional charging systems remain relatively limited and inconsistent throughout the domain capability profiles. Given that EVs are expected to emerge as the largest sources of flexible demand and distributed storage on the electric system, the capability profiles would benefit from clearer and more consistent recognition of managed charging, bidirectional charging, EV aggregators, mobile storage

resources, and EV-enabled grid services as integral elements of a highly flexible, bidirectional electric power ecosystem.

3. Emphasize the Role of EVs to Address Affordability Through Flexible Grid Utilization

VGIC supports the domain capability profile's repeated recognition that affordability must remain a core objective of the GOTF effort. The pre-read document appropriately identifies affordability as one of the primary drivers for enabling flexibility at scale. However, VGIC believes the framework could place even greater emphasis on VGI as a foundational affordability strategy.

Across the domain capability profiles, the narratives recognize the importance of flexible resources, non-wires alternatives, probabilistic forecasting, and co-optimized planning. These concepts are important because they represent a shift away from traditional "worst-case" infrastructure planning assumptions toward more dynamic and operationally optimized approaches.

Transportation electrification is particularly important in this context because unmanaged EV charging has the potential to increase localized peak demand, accelerate infrastructure upgrades, and increase overall system costs, while managed charging and bidirectional charging systems can significantly reduce or even eliminate many of those same costs through flexible load management and grid services. As a result, the distinction between unmanaged charging and the flexible capabilities enabled by EVs and bidirectional charging systems becomes increasingly central to future affordability, system planning, and grid modernization outcomes.

VGIC recommends that the domain capability profiles more explicitly recognize the opportunity to increase utilization of existing infrastructure using VGI solutions before deploying traditional upgrades. Several sections already point in this direction. For example, E1 recognizes the importance of incorporating probabilistic planning, DER flexibility states, and scenario-based forecasting approaches. Similarly, E2 discusses flexible interconnections, scheduled DER operation, and dynamic operating approaches. These concepts collectively support a more dynamic grid architecture capable of accommodating greater levels of variable sources of generation and electrification without relying exclusively on traditional infrastructure expansion.

VGIC recommends that the domain flexibility profiles explicitly identify managed charging and bidirectional charging systems as important tools for avoiding or deferring distribution system investments. This is especially relevant given the increasing emphasis throughout the profiles on



feeder-level forecasting, hosting capacity analysis, and localized operational constraints. EV flexibility can help utilities manage feeder peaks, reduce transformer loading, improve renewable integration, and increase hosting capacity utilization in ways that directly support affordability objectives.

VGIC also recommends greater emphasis on flexible service connections and dynamic operating limits as alternatives to traditional upgrade approaches. The domain capability profiles already acknowledge that flexibility-aware planning and operations will become increasingly important as electrification and DER penetration increase. Building upon this concept, the capability profiles should more directly recognize that operational flexibility can serve as a planning asset capable of unlocking existing grid capacity.

Finally, VGIC recommends a stronger linkage throughout the domain capability profiles between flexibility, affordability, and long-term decarbonization goals. The overarching framework of this effort correctly recognizes that the GOFT must simultaneously support reliability, affordability, electrification, and clean energy integration. However, the profiles could more explicitly emphasize that VGI strategies may be one of the most important mechanisms for achieving all of these objectives simultaneously.

4. Conclusion

VGIC appreciates the significant effort that the DPS, NYSERDA, and the broader stakeholder community have invested in developing the GOTF Domain Capability Profiles and establishing a comprehensive framework for the evolution of New York's electric power ecosystem. VGIC believes that transportation electrification and vehicle-grid integration can play a central role in advancing the domain capability profiles' ultimate outcomes related to flexibility, affordability, reliability, and decarbonization. We look forward to continuing to work collaboratively with DPS, NYSERDA, utilities, industry stakeholders, and other participants in the GOTF proceeding to help realize the vast and growing capability of EV as flexible grid resources supporting a cleaner, more resilient, and more affordable electric system for New York.